```
model = MyNet()
```

That’s all you need to know about PyTorch for now to be productive with it. We will discuss a few other bells and whistles as we progress through the book.

2.5. Solving contextual bandits

We’ve built a simulated environment for a contextual bandit. The simulator includes the state (a number from 0 to 9 representing 1 of the 10 websites in the network), reward generation (ad clicks), and a method that chooses an action (which of 10 ads to serve). The following listing shows the code for the contextual bandit environment, but don’t spend much time thinking about it as we want to demonstrate how to use it, not how to code it.

Listing 2.9. Contextual bandit environment

```
class ContextBandit:
    def __init__(self, arms=10):
        self.arms = arms
        self.init_distribution(arms)
        self.update_state()

    def init_distribution(self, arms):                    1
        self.bandit_matrix = np.random.rand(arms,arms)

    def reward(self, prob):
        reward = 0
        for i in range(self.arms):
            if random.random() < prob:
                reward += 1
        return reward

    def get_state(self):
        return self.state

    def update_state(self):
        self.state = np.random.randint(0,self.arms)

    def get_reward(self,arm):
        return self.reward(self.bandit_matrix[self.get_state()]
[arm])
```